

Solvency Field Theory V12.12

Locally Real Particle Mechanics: Open Energy, Closed Mass, and Settlement

SFT Research Notes

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V12_12_PARTICLE_PROGRAM_DRAFT_READY
REALITY_LOCKED_OPEN_CLOSED_ENERGY_SYNTHESIS
R159_R185F_DERIVATION_INTEGRATED

Abstract

SFT V12.12 is the particle-program companion to the V12.11 reality-locked foundation. It consolidates the R159–R185F audit chain into one locally real particle doctrine. The central claim is that the fundamental particle object is real energy moving at c . When that energy remains open and aimed outward, it is light. When the same energy closes into a self-maintaining orbit, it is mass. A photon has no internal maintenance clock because it is not closed. A massive particle ticks because the closed orbit repeats. No-overwrite is not spatial self-avoidance; it is the permanence of a completed physical settlement write. Formal terms such as causal edges, ledgers, fibers, and vertices are retained as mathematical blueprints only when they translate back into real energy, real motion, real settlement, and real no-overwrite.

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1 The physical picture

The post-shatter universe is treated as an ocean of real energy moving at c . At the most primitive level, this energy is open: it propagates outward as light. It does not stop, rest, or wait. Energy at c can do only two things:

1. remain open and propagate outward;
2. close into stable self-maintaining circulation.

The first state is the photon. The second state is the massive particle.

The physical movie is:

energy ocean -> compactness fluctuations -> self-settlement/focusing
-> bending of c -speed energy -> closure -> stable modes -> particles
-> time, gravity, and expansion

The purpose of V12.12 is to state this picture without letting the formal scaffolding replace the real object. The real object is not a graph edge, not a database record, and not an abstract label in a fiber bundle. The real object is energy moving at c .

2 Energy is gravity is compactness

The unifying SFT reality principle is:

energy, gravity/settlement demand, and compactness
are three readings of one physical condition.

This is a reality-layer principle, not yet a completed microscopic field equation. It says that energy does not merely possess gravity as a separate add-on. Energy is the source of settlement demand, and compactness is the local intensity of that demand.

Mechanically:

- low energy / long wavelength: spread out, low compactness, weak settlement demand;
- high energy / short wavelength: concentrated, high compactness, stronger settlement demand;
- closed energy: compact enough to maintain a stable orbit, experienced as mass.

SFT therefore treats particles, black holes, and the universe as different regimes of one geometry: energy organized by compactness and no-overwrite.

Claim boundary. V12.12 does not assert that ordinary weak-field Newtonian self-gravity with macroscopic G alone binds elementary particles. R183 showed that a naive ordinary-gravity calculation is far too weak at elementary masses. The claim is stronger and more SFT-specific: what appears as gravity at large scales is the large-scale face of the same energy/compactness/settlement condition that must be written microscopically in future SFT work.

3 Definition and branch states

The V12.12 particle definition is:

A particle is real energy moving at c , either open as light or closed into a self-maintaining orbit as mass, leaving permanent settlement history as it writes.

The two branch states are:

- **Photon/light:** real energy at c aimed outward; open branch; no internal maintenance clock.
- **Mass/matter:** same real energy at c aimed around itself; closed branch; self-maintaining orbit; internal maintenance cadence.

The branch distinction is topological and mechanical, not a change of substance. There is one kind of stuff: energy at c . Light is open energy. Mass is closed energy.

4 The photon: open branch

A photon is real energy at c in the open outward state. It is not unreal, not merely a wavefunction, and not merely a mathematical null line.

Its energy and wavelength obey

$$E = \frac{hc}{\lambda}.$$

The locally real reading is:

- higher photon energy: shorter wavelength, greater compactness, stronger settlement/gravity demand, closer to closure threshold;
- lower photon energy: longer wavelength, lower compactness, weaker settlement/gravity demand, farther from closure threshold.

R181 established the clock correction. For geometric-optics phase,

$$\Phi = k_\mu x^\mu,$$

and a ray tangent proportional to k^μ ,

$$\frac{d\Phi}{d\lambda} = k_\mu \frac{dX^\mu}{d\lambda} \propto k_\mu k^\mu = 0.$$

Thus the photon does not experience optical phase ticks along its own null path. An observer measures

$$\nu_{(u)} = -\frac{p_\mu u^\mu}{h},$$

but that is an observer-relative wavefront crossing rate, not the photon's internal clock.

Claim boundary. V12.12 does not claim a universal hard photon radius. The physical support/localization law of light remains open. The photon is real energy with compactness and stress/settlement demand, but its transverse support is not yet derived.

5 The massive particle: closed branch

A massive particle is the same real energy at c folded into a closed, self-maintaining orbit.

For the fermion candidate branch, R176 locked the spinor-Compton geometry:

$$r_{\text{spin}} = \frac{\hbar}{2mc},$$

$$2r_{\text{spin}} = \frac{\hbar}{mc},$$

so the reduced Compton length is the orbit diameter, not the orbit radius.

The branch also has 4π spinor closure in one Compton period:

$$T_{4\pi} = T_C.$$

The one-speed budget is

$$v^2 + u_{\text{internal}}^2 = c^2.$$

At rest, the energy is aimed internally around itself. As spatial motion increases, more of the c -budget is allocated to propagation and less to internal cadence, giving time dilation.

The massive particle's clock is not added from outside. The clock is the closure cadence of the real closed energy.

6 Formation: why energy becomes particles

Energy at c cannot stop. When compactness rises, the settlement/gravity response bends its path. If the localized condition is strong enough, open energy can be rerouted into closed circulation. Stable closure is mass.

The mechanical formation chain is:

open energy -> sufficient compactness -> focusing/settlement
-> bending -> closure -> stability test -> persistent particle

R183 supplies an important conservation guardrail: one isolated free photon cannot spontaneously become one isolated massive particle in empty space. A single null four-momentum satisfies $p^2 = 0$, while a massive particle satisfies $P^2 = m^2 c^2 > 0$. If no other edge, field, or outgoing channel carries momentum, conservation forbids the transition.

For two null inputs,

$$M^2 c^4 = 2E_1 E_2 (1 - \cos \theta).$$

Parallel inputs remain null. Non-collinear inputs can create a timelike invariant. Thus formation requires a physical encounter, localized energy condition, or interaction vertex.

Once invariant mass and spin are formed, a thin circular c -speed core satisfies

$$J = \frac{E_{\text{core}} r}{c}.$$

For $J = \hbar/2$ and $E_{\text{core}} = mc^2$,

$$r = \frac{\hbar}{2mc}.$$

The lag/settlement envelope does not need to invent this radius. It must stabilize it. R183 derived the envelope targets:

$$E_{\text{env}}(r_0) = 0, \quad E'_{\text{env}}(r_0) = \frac{Jc}{r_0^2},$$

$$E''_{\text{env}}(r_0) > -\frac{2Jc}{r_0^3}.$$

These are necessary stability conditions for a native settlement/focusing action. The action itself remains open.

7 Energy addition and removal

For a photon, changing energy changes compactness:

- added energy shortens wavelength, increases compactness, and strengthens settlement demand;
- removed energy lengthens wavelength, decreases compactness, and weakens settlement demand.

For a massive particle, small added energy does not smoothly change its rest mode. The internal closure is rigid. Ordinary added energy goes into spatial motion:

- small added energy: kinetic motion, faster spatial propagation, slower internal clock;
- removed kinetic energy: slower spatial propagation, faster internal cadence;
- large localized collision energy: new timelike invariant, new closure, pair production or particle creation.

This is the locally real version of quantization: stable closed energy modes are discrete.

8 No-overwrite corrected

The old intuitive phrase was:

the helix cannot cross its own past spatial path.

R182 showed this cannot be the fundamental rule. A continuous path has neighboring points arbitrarily close to its past, and the closed massive orbit must revisit the same spatial projection every cycle.

The frozen V12.12 rule is:

a completed physical write leaves permanent settlement history,
so the same full event cannot be written again.

Spatial recurrence is allowed. History recurrence is not.

The same coordinates can be occupied later because it is not the same physical event. The old write remains settled. The new write requires new capacity. Expansion is the large-scale expression of this capacity growth.

9 Time, gravity, and expansion

The particle program is not isolated from the cosmological program. It supplies the local machinery for the same process that appears globally.

At particle scale:

closed energy repeats -> maintenance cadence -> proper time

At regional scale:

compact energy settles geometry -> gravity/focusing

At cosmic scale:

writes accumulate -> boundary capacity grows -> expansion

Thus the same rule has three faces: time, gravity, and expansion. This is why V12.12 belongs under V12.11. V12.11 gives the reality-locked foundation and master action guardrails. V12.12 states what the action is acting on.

10 Reality-lock translation table

formal term	locally real translation	meaning	guardrail
causal edge	real energy path	The actual physical path of energy moving at c.	Blueprint allowed; ontology is energy.
null branch	open outward energy	Photon/light branch.	No internal clock.
timelike branch	closed self-maintaining energy	Mass/matter branch.	Internal maintenance cadence.
ledger address	settled physical event	Full physical write into permanent history.	Not a database key.
injective write map	same completed write cannot happen twice	No-overwrite stated precisely.	Same spatial projection may recur.
fiber	transported circulation/geometry property	Math for spin/charge/helicity behavior.	Not an abstract decoration.
SU(2) fiber	physical half-turn / 4pi spin geometry	Fermion spinor closure.	Representation layer only.
U(1) connection	frame/holonomy/circulation orientation	Charge sector candidate.	Charge quantization open.
lag envelope	settlement/focusing response	Physical envelope around trapped energy.	Native action open.
vertex	physical encounter/rerouting event	Where open energy can form closed modes, scatter, absorb, emit.	Must obey conservation constraints.

11 The 6π lineage

The V12.12 particle program was triggered by a 6π open from the mass-hierarchy bridge.

The retained values are:

$$D = 17877.208689571427,$$

$$K_{\text{req}} = 18.890588898087,$$

$$\beta_{\text{req}} = 0.300130654930242,$$

$$6\pi = 18.849555921539.$$

The difference is

$$K_{\text{req}}/(6\pi) - 1 = 0.002176867015802,$$

or

$$+0.217686702\%.$$

V12.12 preserves the honest status: 6π remains a strong local closure-load candidate, but the 6π -to- K mechanism is not derived.

The failed routes are retained:

- 6π as ordinary tube volume: rejected;
- 6π as three independent canonical action payments: rejected;
- 6π as forced closure-vector scalar: possible candidate, not uniquely derived;
- 6π -to- K : open.

12 Current family status

V12.12 does not claim a complete particle-by-particle species derivation.

What is currently supported:

- selected-family rigidity is conditionally supported under legal variations;

- fermion branch mechanics are supported as a candidate branch;
- stable closure modes are treated as discrete fixed points of no-overwrite/support/focusing machinery;
- full observed spectrum is not exhaustively assigned here.

13 Claim boundary matrix

topic	claim	status	boundary
Primary ontology	The fundamental object is real energy moving at c.	FROZEN	Does not define energy microscopically beyond locally real SFT primitive.
Photon	Open outward energy at c; no internal maintenance clock.	FROZEN	Physical support/localization law remains open.
Photon energy	$E=hc/\lambda$; higher energy means shorter wavelength and greater compactness.	FROZEN_INTERPRETIVE	The exact transverse compactness law is not derived.
Massive particle	Same real energy closed into a self-maintaining orbit.	FROZEN	Native nonlinear settlement/eigenvalue equation remains open.
Fermion branch	$r_{\text{spin}}=\hbar/(2mc)$, diameter= $\hbar/(mc)$, 4pi closure in one Compton period.	FROZEN_BRANCH	Not yet a complete species map.
No-overwrite	Completed physical settlement write cannot be repeated.	FROZEN	Not spatial self-avoidance.
Energy=gravity=compactness	SFT treats energy, settlement/gravity demand, and compactness as three readings of one physical condition.	REALITY_LAYER_PRINCIPLE	Not yet a completed covariant microscopic equation.
Formation	Energy self-settles; at sufficient compactness open energy can close into stable modes.	WORKING_MECHANICAL_PRINCIPLE	Simple isolated photon self-capture is forbidden; formation requires interaction/localized invariant.
Nine modes / families	Stable closure modes are discrete fixed points of the no-overwrite/slot/focusing machinery.	CONDITIONAL	Full particle-by-particle species assignment not complete.
6pi	6pi remains a strong local closure-load candidate near K_{req} .	PRESERVED_OPEN	6pi-to-K mechanism not derived.
Dark sectors	No new dark matter/dark energy substances are introduced in V12.12.	GUARDRAIL	Predictive tests still required.

14 Equation index

id	equation	meaning	source
photon_energy	$E = h c / \lambda$	Photon energy-wavelength relation; compactness guide.	Photon section
null_phase	$d\Phi/d\lambda = k_{\mu} dX^{\mu}/d\lambda$ proportional to $k_{\mu} k^{\mu} = 0$	Photon has no internal optical phase tick along its own null ray.	R181
observer_frequency	$\nu(u) = -p_{\mu} u^{\mu} / h$	Observed wavefront crossing rate; not photon proper clock.	R181
fermion_radius	$r_{\text{spin}} = \hbar/(2 m c)$	Fermion closed-orbit centerline radius.	R176
fermion_diameter	$2 r_{\text{spin}} = \hbar/(m c)$	Reduced Compton length as orbit diameter.	R176
one_speed_boost	$v^2 + u_{\text{internal}}^2 = c^2$	Spatial and internal c-budget split.	R176
compton_clock	$\nu_C = m c^2 / h$	Massive maintenance cadence.	R180/R185F
spin_closure	$\alpha_{\text{spin}} = 4 \pi N_C$	4pi spinor closure on fermion branch.	R176/R180

id	equation	meaning	source
formation_invariant	$M^2 c^4 = 2 E_1 E_2 (1 - \cos \theta)$	Two non-collinear null inputs can create timelike invariant.	R183
core_angular_momentum	$J = E_{\text{core}} r / c$	Thin circular c-speed core identity.	R183
env_stability_1	$E_{\text{env}}(r_0)=0$	Envelope preserves exact spin radius.	R183
env_stability_2	$E_{\text{env}}'(r_0)=Jc/r_0^2$	Radial equilibrium target.	R183
env_stability_3	$E_{\text{env}}''(r_0)>-2Jc/r_0^3$	Positive second-variation target.	R183
selected_D	$D = 17877.208689571427$	Selected family rollup value.	R172
K_req	$K_{\text{req}} = 18.890588898087$	Hierarchy bridge requirement.	R172
beta_req	$\beta_{\text{req}} = 0.300130654930242$	Alternative beta bridge requirement.	R172
sixpi	$6 \pi = 18.849555921539$	Closure-load candidate.	R175-R179

15 Open problems

open problem	question	priority	status or next lane
photon sup- port/localization law	What physical transverse/support structure does a photon have?	HIGH	Do not claim zero thickness; sup- port law open.
microscopic settlement equation	What covariant local/nonlocal equation realizes en- ergy=gravity=compactness?	HIGH	Next-era particle dynamics.
formation/capture dy- namics	What exact vertex conditions close open energy into a massive branch?	HIGH	R183 gives constraints, not ampli- tudes.
mass eigenvalue deriva- tion	How do nine/family modes map to the observed spectrum?	HIGH	Structured mass lane.
charge quantization	Why those charge signs and frac- tions?	HIGH	Not solved by V12.12.
radiation capacity source	How does open energy source boundary capacity without a clock?	HIGH	Cosmology/CMB lane.
6pi-to-K mechanism	Why should closure load map to K_{req} ?	MEDIUM_HIGH	Preserved open.
SPARC prediction lane	Does h_lock/morphology improve rotation predictions?	EMPIRICAL	Run after particle document.

16 Source rollup

source	topic	mined result	particle program use
R156-R159	Strict K vs beta invariance audit	Neither K nor beta is globally in- variant under unrestricted pertur- bations.	This forced legal/destructive vari- ation separation.
R160-R172	Selected-family reconstruction and legal perturbation closure	$D=17877.208689571427$; $K_{\text{req}}=18.890588898087$; $\beta_{\text{req}}=0.300130654930242$.	The selected family is condi- tionally rigid under legal/gauge- equivalent variations.
R173	Dual survival gate	A candidate particle must be both admissible to settled history and structurally stable.	Mechanical failure modes are crash or fly apart.
R174-R176	Compton/spinor branch	Fermion candidate: $r_{\text{spin}}=\hbar/(2mc)$, diame- ter= $\hbar/(mc)$, 4π closure in one Compton period.	Massive particle = closed c-speed energy orbit.
R177-R179	6pi action route audits	Tube-volume and triple indepen- dent action-payment routes re- jected; closure-vector load sur- vives as candidate.	6pi remains open as mechanism.
R180	Base/fiber formal cleanup and theta quarantine	Bare bulk theta phase quaran- tined; spin and charge as trans- ported properties over base his- tory.	Formal architecture is blueprint only after R184R.

source	topic	mined result	particle program use
R181	Null branch audit	Photon has no internal clock; optical phase is constant along its own ray.	Photon is real open energy, not a clockless abstraction.
R182	Support and no-overwrite correction	Hard spatial tube self-avoidance rejected; core/envelope architecture selected.	No-overwrite is permanent settlement history, not spatial nonintersection.
R183	Formation and stability targets	Single isolated photon cannot self-capture; formation needs interaction/localized invariant; stability targets derived.	Mass formation requires a physical formation/closure event.
R184R	Locally real reality lock	Real energy at c restored as ontology; graph/fiber language demoted to blueprint layer.	The particle is a real machine, not a data structure.
R185F	Foundation freeze	V12.11 reality-locked foundation frozen and prediction-ready.	V12.12 particle document is the companion doctrine.

17 Final frozen statement

The fundamental object is real energy moving at c .
 Open energy is light. Closed energy is mass.
 Closed energy creates cadence, which is time.
 Compact energy creates settlement, which is gravity.
 Permanent writes require capacity growth, which is expansion.
 No-overwrite is the permanence of every completed physical write.